2026 Moxbridge Interviews Questions and Answers[1]

November 9, 2026

Contents

1 The Prisoners and the Lever **2**

2 The Wonky Chessboard **7**

3 The Lagrange Interpolating Polynomial **14**

4 Comparing and Estimating Numbers **16**

5 Designing a Dam **17**

6 Triangles In A Cube **18**

7 Mean of Neighbours **20**

8 Spaced Apart In A Grid **21**

9 25 Horses **23**

10 Sketch of a Graph **24**

11 Box From a Rectangle **25**

12 Leaky Jug **26**

13 Centre of Mass of a Plane **28**

14 Iteratively Merging a Set **29**

15 The Ball and the Boat **30**

16 Three Pegs **34**

[1]The source code for this document is available at `https://github.com/HenryGinn/Moxbridge`

1. There are $n+1$ prisoners who have been captured by an evil logician. Each prisoner is put into individual cells. One at a time, in no particular order, each prisoner will be brought into a room that has a lever in where they are given a choice of whether to change its state or not. The prisoner will then be sent back into their cell and the next prisoner brought in to the room with the lever. This process repeats until one of the prisoners announces that all prisoners have been in the room. If they are correct then all of the prisoners are free to go, although if they are wrong then the prisoners are imprisoned forever/sentenced to death.

   The following may be assumed.

   1. The lever has two states, up and down.
   2. The prisoners are allowed to communicate freely before being put into their cells to agree a strategy. In particular, they know how many prisoners there are.
   3. At any point in time for any prisoner, there exists a future point where they will be brought into the room again (unless an announcement is made before this occurs). We call this the recurrent constraint.
   4. Standard evil logician rules apply (no funny business!).

   (a) If the order that the prisoners enter the room is random, define a strategy as "Certain" if there is a probability of 1 that it succeeds. Prove that waiting a really long time and then guessing is not a certain strategy, independent of how long the wait is.

   Hints:

   - It is not necessary to compute the probability that all prisoners have been in the room by a certain point.
   - It is sufficient to show that the probability of failure is bigger than some positive number.
   - To better understand the definition of certainty, note that having a probability of 0 of failure does not mean that a strategy will never fail - this is only true if the sequence of prisoners is bounded in length.

   Solution: For a strategy to fail, all we need is that the probability that a given prisoner has not entered the room at a given point is positive. The probability that any given prisoner is brought into the room with the lever on a given time step is $p = \frac{1}{n+1} < 1$. This means that the probability that they are not brought into the room on the first N time steps if $(1-p)^N$. In the limit as $N \to \infty$, this is 0, but with the "wait a long time and guess" strategy, N must be some finite number. A non-zero real number raised to a natural number is never 0. The probability that there exists at least one prisoner who has not been brought into the room with the lever by the Nth visit is at least as big as the probability that one given prisoner has not been brought into the room. Therefore, for any value of N, we have bounded the probability away from 0 so we are done.

(b) Find a strategy when all of the prisoners know that the lever starts in the up position.

Hints:

- The announcement that all prisoners have visited the room does not need to happen at the earliest possible moment. The only requirement is that all prisoners have been in the room at least once, meaning all prisoners may have visited the room multiple times when the announcement is made.
- Whether any individual prisoner has visited the room is not important, only the total number.
- Only one prisoner needs to be able to make the claim that all have visited the room.
- If a prisoner flicks the lever and another prisoner flicks the lever again before the prisoner making the claim enters the room, this state will be indistinguishable to where no prisoner flicked a lever.

Solution: This strategy works via communicating all information to a special prisoner who will be referred to as the communicator. All other prisoners will be referred to simply as prisoners. The key is to ensure that the communicator counts each prisoner at least once, but no more. Each prisoner follows a simple rule: if the lever is up and they have not flicked it before, flick the lever down, but otherwise do nothing. If the communicator enters the room and sees the down, flick it back up, but otherwise do nothing.

To understand this, consider the information that the communicator can deduce depending on what state they see the lever in. The lever being down means a prisoner who had not flicked the lever before has flicked the lever. When the communicator flicks the lever back up they reset it back into the original state, and they add one to the count of prisoners who have flicked the lever.

As no prisoner can flick the lever twice, no prisoner will get double counted. As no one can flick the lever up to reset the lever other than the communicator, no prisoner can get missed. As the communicator will know how many prisoners there are due to being allowed to communicate beforehand, they will know when all prisoners have been in the room when they have counted to n. The existence of the communicator is why this problem was defined with $n + 1$ prisoners instead of n.

(c) Define a strategy as "Robust" if it works even when a malicious chooser decides the order that the prisoners enter the room. Prove that a robust strategy is always certain. Is your answer to part b robust?

Solution: We consider the contrapositive statement - if a method is not certain then it is also not robust. We need to show the existence of a sequence of prisoners that causes the prisoners strategy to fail. A malicious chooser could give the same sequence, therefore making the strategy not robust. Suppose that a strategy was certain but there was no such sequence that causes the prisoners' strategy to fail. This would imply that the strategy worked 100% of the time which contradicts the fact that the strategy was not certain.

We know that the "wait a long time and choose" strategy from part a is not robust as it is not certain, but we can use it to build intuition about robustness. This strategy fails because the malicious chooser knows the prisoners have to make an announcement after some amount of time. Suppose one of the prisoners announces after entering the

room 10,000 times, yet one of the prisoners had not entered the room once. The prisoners complain that the malicious chooser had broken the recurrent constraint, but the malicious chooser claims they planned to put the last prisoner in the room the next time. The prisoners' strategy means they must make an announcement within a finite amount of time. The malicious chooser will always have the defense that the prisoners just had not waited long enough.

The solution offered to part b is not susceptible to the same attack however. There is no limit on the amount of time that the communicator can wait until they make an announcement. If the malicious chooser refuses to put at least one of the prisoners in the room after some point, they would have broken the recurrent constraint. If there are k prisoners who have not flicked the lever after t room visits, then the malicious chooser must put at least one of these prisoners in the room after this point due to the recurrent constraint. This will occur after a finite amount of time and would lead to one of the k prisoners flicking the lever. By the same reasoning, the communicator must subsequently be put into the room with the lever. There would then be $k-1$ prisoners who had not entered the room and this process repeats by induction until all prisoners had been in the room with the lever.

(d) Suppose that none of the prisoners know what position the lever starts in. Prove whether or not your strategy from part b works. If it does not, find a working strategy given this additional constraint and prove whether or not your strategy is robust. Can this strategy be modified to work if a malicious actor is also allowed to flip the lever's state up to k times where k is known to the prisoners?

Hints:

- The strategy given as a solution to part b does not work. If the communicator enters the room first and sees the lever down, they will not know whether this is because they are the first in the room or if another prisoner has been in before and flicked it down. If the communicator counts to $n-1$ then either the lever started down or they have not waited long enough.
- The main idea from the part b of counting prisoners by flicking the lever down can be extended.
- Consider letting prisoners flick the lever multiple times.
- Use the pigeonhole principle.

Solution: This strategy works using the pigeonhole principle. The strategy for each prisoner is updated so that they will flick the lever down if they have flicked the lever at most once. The communicator does the same as before, flipping the lever back up to reset it and adding 1 to a count. If the lever starts in the up state, the communicator will count the lever being in the down state $2n$ times when all n prisoners have flicked the lever down two times each.

If the lever started in the down position however, then it is possible for the prisoners to have collectively only flicked the lever down $2n-1$ times yet the communicator has seen the lever being down $2n$ times. This occurs when the first time the communicator enters the room the lever had not been flicked down by any prisoner, yet it started down so was counted anyway.

If there were only $n-1$ prisoners, they could collectively only flick the lever down $2n-2$ times. Even if the lever started in the down position, the most the communicator could count to would be $2n-1$, making it impossible for the communicator to count the lever being in the down position $2n$ times. For the communicator to count to $2n$ each of the n prisoners must have been in the room at least once by the pigeonhole principle. In fact at most one prisoner could not have been in the room twice. This strategy is robust by the same argument given in part c but this time k starts at $2n$ instead of n.

Consider the case where the lever is known to start in the up position and the malicious chooser can flip the state of the lever a single time. This situation is very similar to not knowing the starting state of the lever without a malicious actor interfering. The difference is the timing of when the lever state changes, but this does not affect our solution. In fact we could even assume that the lever starts in the up position and consider a malicious actor who interferes up to $k+1$ times instead. We can use this insight to modify our strategy to work when the malicious chooser flips the lever up to k times.

Instead of flipping twice and counting up to $2n$, prisoners flip α times and the communicator counts up to β where α, β are to be determined. When the communicator makes the announcement, the others prisoners have flipped the switch at least $\beta - k$ times and at most $\beta + k$ times. If we make β too high then it may not be possible for the prisoners to flick the lever enough for β to be reached. This gives equation (1a). If β is too low then our pigeonhole argument will not work as it could be possible for not all the prisoners to flick the lever yet β is still reached. This gives equation (1b).

$$\beta \leq n\alpha - k \tag{1a}$$

$$(n-1)\alpha + k \leq \beta \tag{1b}$$

Solving these inequalities gives $\alpha \geq 2k$. To minimise the length of time the prisoners are flicking levers, we choose α as small as possible and get $\alpha = 2k$. Substituting into either equations in (1) gives $\beta = k(2n-1)$. Note that this solution is also robust.

(e) Find a strategy that is certain where the initial state of the lever must also be determined. Prove whether or not this strategy is robust.

Hints:

- First determine how it is possible for the communicator to learn the state of the initial lever. There are two such cases to consider.
- If the prisoners only flick the lever down, then when the lever is down the communicator can know whether this is due to them or a prisoner on all but their first visit to the room.

Solution: We consider two cases. Suppose prisoners can flip the lever before the communicator. The difficulty here is that the prisoner who flips would need to deduce that they are the first to flip and then signal the lever state to the communicator. Alternatively the communicator could always flip the lever first which has the advantage that they would know the starting state had been preserved. Here the difficulty is that the other prisoners must flip the lever at some point and the communicator now needs to send some sort of signal to the other prisoners that they are allowed to flip the lever.

We explore the latter. The rule for the prisoners is that once they have seen the lever in both states, the lever is up, and they have not flicked it before, then flick it down[2]. The rule for the communicator is more complicated. If they leave the room with the lever flicked up and the next time they enter they find it down, then they add one to their count. They will not announce until this count reaches n as before, and by the logic of easier version this is a guarantee that all prisoners have been in the room.

The complication is that the communicator will need to sometimes pull the lever down themselves. If they have left the lever up for a long time and it is not getting flicked down, then it would indicate that either a prisoner has only seen the lever in the up position, and thus has not received the signal to pull the lever down, or that prisoner has not been in the room. The communicator would flick the lever down and leave it for a while to give the other prisoners an opportunity to see it, and then flick it back up to give them an opportunity to flick it down.

This strategy is not robust to malicious choosers. Suppose the lever starts in the up position and all prisoners see it. The communicator flips it down, a prisoner sees it down and flips it back up, and then the communicator enters the room again. They add one to the count and flip the lever down. The malicious chooser then repeatedly puts either the communicator or the prisoner who already flipped the lever back into the room. Eventually the communicator infers that the prisoners have not received the signal, so flicks the lever up. The malicious chooser can avoiding breaking the recurrent constraint at this stage. Eventually the communicator flips the lever back down again and the cycle repeats.

If the prisoner order was random however, then there is a non-zero chance, p, that a given prisoner gets the signal that they can flick the lever and sees it in the up state. As $1-(1-p)^n \to 0$ as $n \to \infty$, the chance that the game goes on forever without the prisoner getting the opportunity to flick the lever is 0. There are a finite number of prisoners, therefore there is a probability of 1 of success.

[2]Seeing both states means someone must have flipped the lever, and as lever flipping cannot start until the communicator has flipped the lever, they know it must be safe to start flipping the lever.

2. Take an arbitrary convex quadrilateral, $ABCD$. Divide each edge into n sections of equal length and connect opposite points to form a grid as shown in figure 1. Consider regions where there is exactly one region in each row and in each column - we will refer to this as a rook configuration[3]. Regions are shaded such that they form a rook configuration. The ultimate aim of this question will be to show that the area of the quadrilateral $ABCD$ is n times larger than the total area of the shaded regions. Whilst you are free (and encouraged) to attempt this question at this stage, the following series of questions guide you through a full proof.

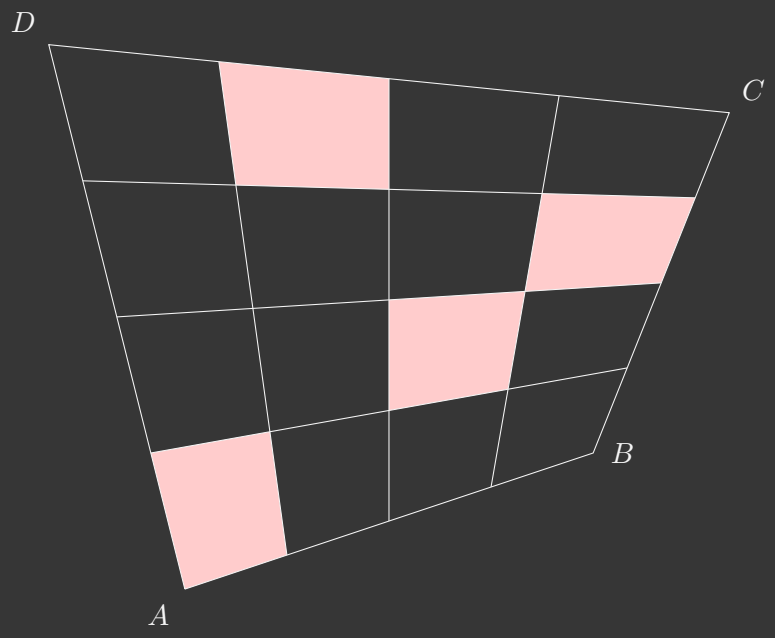


Figure 1: The wonky chessboard

(a) Consider the $n = 2$ case shown in figure 2. Here we have defined the midpoints of AB, BC, CD, and DA as P, Q, R, and S respectively. Show that the quadrilateral $PQRS$ is a parallelogram.

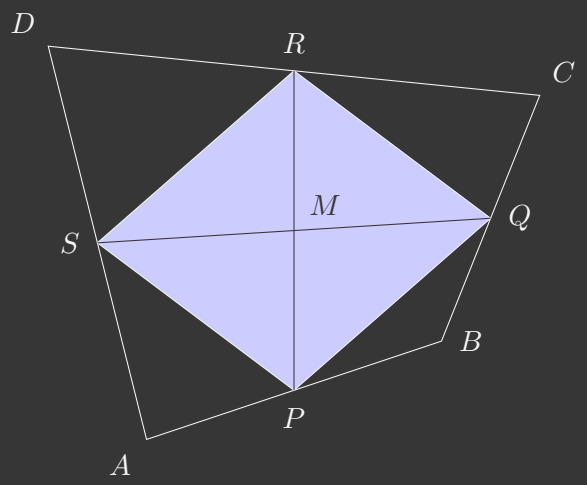


Figure 2: The parallelogram formed by connecting midpoints

[3]These are the arrangements of rooks from chess where no rook could attack any other rook.

Hints:

- How can a parallelogram be characterised?
- Doing a lot of algebra will likely be very messy. What tools do you know of for working with geometry problems where there is a lot of flexibility?
- Travelling in a loop around the quadrilateral results leads you back to where you started - can you use this to form an equation?

Solution: It is sufficient to prove that PQ is parallel to SR and that SP is parallel to RQ. The flexibility of a quadrilateral makes this awkward with algebra which motivates using vectors. Define vectors $\vec{a}$, $\vec{b}$, $\vec{c}$, and $\vec{d}$ as AP, BQ, CR, and DS respectively, as shown in figure 3. Note that $2 \cdot (\vec{a} + \vec{b} + \vec{c} + \vec{d}) = \vec{0}$ as this describes a complete path around the quadrilateral, starting and ending in the same location. This is applied in equation 2 which shows that $PQ||SR$ and the proof for $SP||RQ$ follows almost identically.

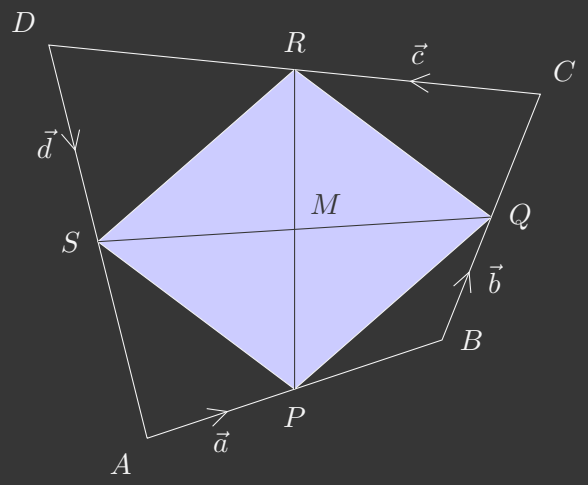


Figure 3: The parallelogram formed by connecting midpoints.

$$PQ = \vec{a} + \vec{b} = \vec{a} + \vec{b} - (\vec{a} + \vec{b} + \vec{c} + \vec{d}) = -(\vec{c} + \vec{d}) = SR \tag{2}$$

(b) Hence or otherwise, show that the result is true for the $n = 2$ case as shown in figure 4.

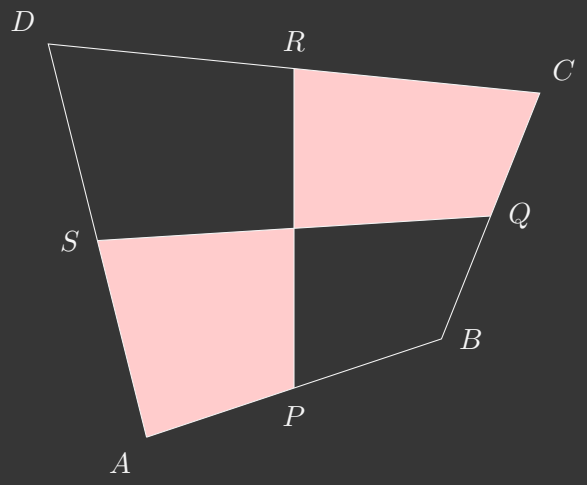


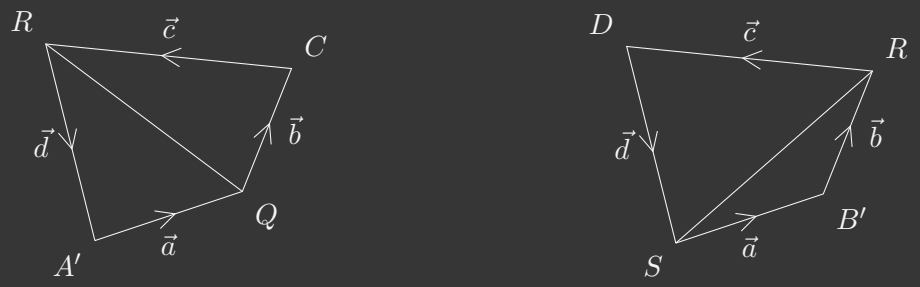
Figure 4: A 2×2 wonky chessboard

Hints:

- Continue with a vector-based approach.
- Split each of the regions into two triangles, one from the parallelogram and one not.

Solution: It is sufficient to show that the red-shaded region has equal area to the non-shaded region in figure 4. Define the intersection of PR and SQ by M. As we know that $PQRS$ is a parallelogram by part a, we can argue by symmetry that the triangles MQR and MSP are congruent, as are MRS and MPQ. The problem therefore reduces to showing the combined area of triangles QCR and APS is equal to the combined area of PBQ and SRD.

If the triangle APS is translated by $\vec{a}+\vec{b}$ (this can be thought of sliding it along an edge of the parallelogram) then SP is mapped onto RQ. The new position of the remaining vertex, A, is labelled as A' and is shown in figure 5a. This forms a quadrilateral, $A'QCR$ with edges defined by $\vec{a}$, $\vec{b}$, $\vec{c}$, and $\vec{d}$ when going anticlockwise. Similarly triangle PBQ can be translated by $-(\vec{c}+\vec{d})$ (sliding along the other edge of the parallelogram) to form $SB'RD$ as shown in figure 5b. This quadrilateral can be seen to be identical to $A'QCR$ as their edges are made of the same vectors in the same order[4], and therefore they have the same area. As these quadrilaterals are composed of the diagonally opposite triangles of interest we are done.



(a) Red corners merged

(b) Non-red corners merged

Figure 5: Sliding opposite corners along the sides of the parallelogram.

(c) Prove that the total area of two regions remains the same when the rows that they occur in are swapped around. That is, prove that $X + Z = W + Y$ in figure 6.

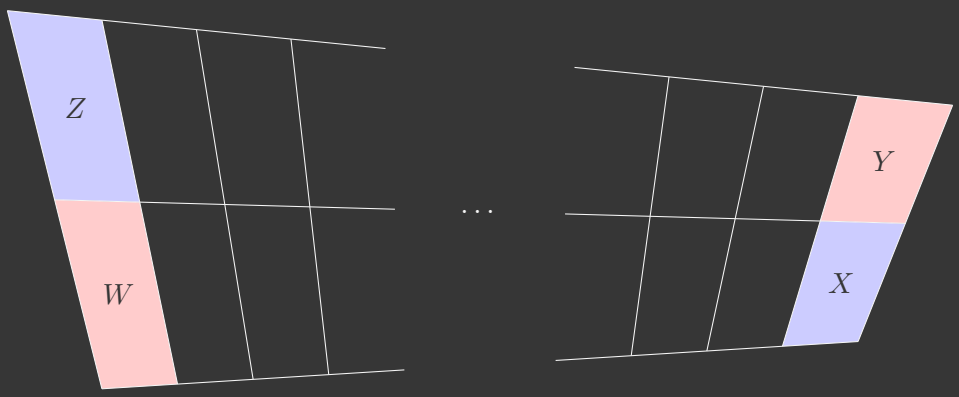


Figure 6: Swapping two rows where highlighted regions are an arbitrary number of columns apart.

[4]This last condition on the order is necessary. Can you think of two quadrilaterals where the edges are made of the same four vectors but the shapes are different?

Hints:

- Use the result for $n = 2$
- The four regions formed by the intersection of two rows with two adjacent columns is an $n = 2$ wonky chessboard.
- Label the regions in a small example of two rows and write out the relationship between the areas for each pair of adjacent columns - how can they be combined in a way that simplifies?

Solution: We apply the result from part b to each adjacent pair of columns. Define the regions along the first row as U_1 to U_n and the regions along the second row as L_1 to L_n, as shown in figure 7. Restated, the problem is to show that the area of the red-shaded area, $L_1 + U_n$, is equal to the blue-shaded area, $L_n + U_1$.

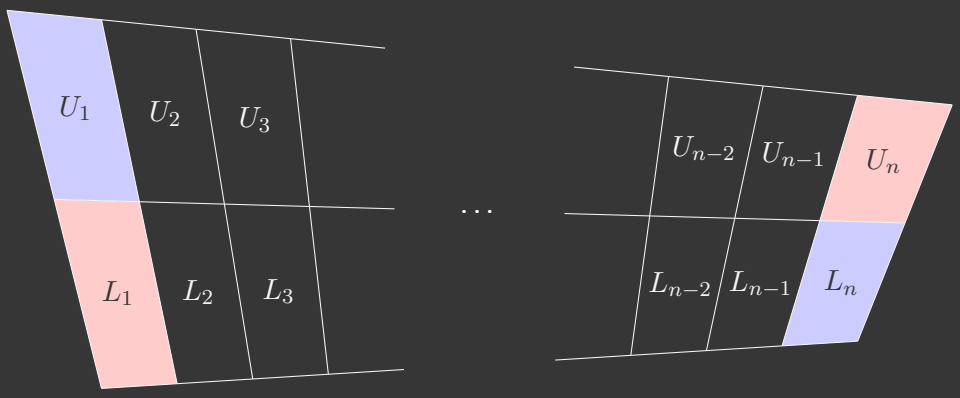


Figure 7: Labelling the regions in figure 6

Applying the result for $n = 2$ to each 2×2 subgrid formed by adjacent columns in figure 7 gives equation (3). The equations on the right hand side telescope when added together leaving $U_1 - U_n = L_1 - L_n$. This is equivalent to $U_1 + L_n = U_n + L_1$ which is precisely the comparison of areas desired.

$$\begin{array}{cc} U_1 + L_2 = U_2 + L_1 & U_1 - U_2 = L_1 - L_2 \\ U_2 + L_3 = U_3 + L_2 & U_2 - U_3 = L_2 - L_3 \\ \vdots & \vdots \\ U_k + L_{k+1} = U_{k+1} + L_k & \implies \quad U_k - U_{k+1} = L_k - L_{k+1} \\ \vdots & \vdots \\ U_{n-2} + L_{n-1} = U_{n-1} + L_{n-2} & U_{n-2} - U_{n-1} = L_{n-2} - L_{n-1} \\ U_{n-1} + L_n = U_n + L_{n-1} & U_{n-1} - U_n = L_{n-1} - L_n \end{array} \tag{3}$$

(d) Deduce from parts b and c that the result is true for all $n \geq 1$.

Hints:

- Any rook configuration can be reached from any other rook configuration by a series of swaps of adjacent rows.
- Show that part c implies that all rook configurations of shaded regions have equal area. The result follows immediately from here.

Solution: The $n = 1$ case is trivial and the $n = 2$ case has already been proven. The idea for larger n is to show that all rook configurations of shaded regions have the same area and then deduce that this area must be n times smaller than the total quadrilateral area by symmetry.

Motivated by the idea that a rook configuration corresponds to a permutation matrix, we use the fact that all permutations of n elements can be achieved by a series of swaps of adjacent elements, known as transpositions. This means the problem reduces to showing that a pair of rows that the shaded regions are in can be swapped. It is sufficient to show that this swapping property is true for adjacent rows where the columns of the shaded regions are not necessarily adjacent. From this the fact that arbitrary rows can be swapped follows from how an arbitrary finite permutation can be achieved by a sequence of transpositions. This was proven in part c.

Now that it has been established that all rook configurations of shaded areas have the same total area then there are two ways of reaching the desired conclusion. For the first method we cover the quadrilateral with an additional $n - 1$ rook configurations such that no two rook configurations share the same regions. As there are n rook configurations of regions which have equal area that cover the whole quadrilateral they must each be n times smaller in area than the whole quadrilateral.

If we are not convinced that the quadrilateral can be decomposed into non-overlapping rook configurations containing the original rook configuration then we can also argue by symmetry. Each rook configuration corresponds to a permutation matrix which in turn corresponds to a permutation of n distinct elements. By symmetry there can be no bias in the number of permutations that move each element to each position. This means that if we ask how many times a given region is shaded as we consider all rook configurations, we will get the same answer independently of what square we choose.

(e) Prove that this result is also true in three dimensions.

Hints:

1. Think how you can reuse your previous results. You likely only need to find an argument for the base case.
2. Consider the fact that the volume of a tetrahedron is a function only of the area of the base and the perpendicular height from the base.

Solution:

The three-dimensional equivalent of an $n \times n$ wonky chessboard has n^2 coloured regions and n^3 total regions meaning we are attempting to prove that the total coloured volume is n times smaller than the total volume. We also note that in two dimensions, any four

points define a convex quadrilateral (as long as no three points lie on the same line), but the analogue in three dimensions is more complicated. If the eight vertices of a cube are moved around, the resulting faces are not necessarily planar (that is, they do not like in a plane). For example, (0, 0, 0), (1, 0, ,0), (1, 1, 0), and (0, 1, 1) do not define a planar quadrilateral. It may be tempting to restrict to the case where all faces are planar, but this turns out not to be necessary and the result is still true for a skew hexahedron (the name of a three dimensional quadrilateral with non-planar faces).

We note that the proof of the general case for the two-dimensional wonky chessboard when assuming the $n = 2$ case can be extended to work for any finite number of dimensions. It is therefore sufficient to prove the $n = 2$ case in three dimensions. Before looking at the three-dimensional case, we return to two dimensions and prove the $n = 2$ case in a different way that will extend more easily to higher dimensions.

Figure **??** shows that the $n = 2$ wonky chessboard decomposed into eight triangles by connecting the four corners to the centre with vertices labelled as in figure **??**. We show that the top two triangles, labelled 1 and 2, have equal area and the result will hold for the other pairs of triangles by symmetry.

If DR and RC form the base of the triangles, then as RM is a shared edge they must have the same altitude. As DR and RC have the same length, the triangles have the same area because the area is proportional to the product of the base and the height. In each pair of triangles, one belongs to a shaded region in FILL ME IN and one belongs to a non-shaded region. This shows that the shaded and non-shaded regions must have the same area.

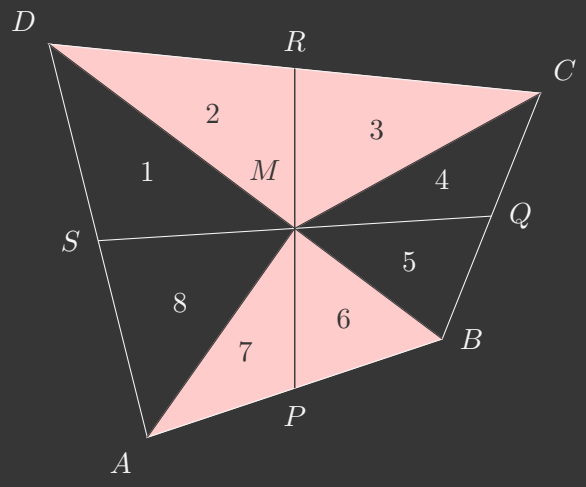


Figure 8: The $n = 2$ wonky chessboard can be decomposed into pairs of triangles of equal area.

We can argue similarly in three dimensions, this time with tetrahedra instead of triangles. Explicitly, we aim to decompose the $2 \times 2 \times 2$ wonky chessboard into pairs of tetrahedra of equal volume, where one of each pair is in a shaded region and the other is not. The volume of a tetrahedron is $\frac{1}{6} \cdot$ Area of base $\cdot$ Perpendicular height meaning we expected 24 pairs of tetrahedra in the whole chessboard[5].

For each of the six faces and for each of the four edges of those faces, an line can be drawn from the centre of the face to the midpoint of the edge. Pick one face and one edge and call that edge the spine. These are shown as dashed lines in figure **??**. The two ends each of the edge that the spine connected to are two corners of the chessboard. The origin, the

end-points of the spine, and one of the corners associated with the spine are four points which define a tetrahedron. Choosing the other corner gives the other half of the pair of tetrahedra.

There are only two tetrahedra that can contain a given spine, so the 24 spines give the 24 pairs of tetrahedra that we expected. These tetrahedra are non-overlapping so must compose the chessboard exactly. By construction, both tetrahedra share the face containing the origin and the spine - we consider this to be the base of each of our tetrahedra. As the spine connects to the midpoint of the edge of the chessboard face, the distance along this edge to each of the corners must be the same. The perpendicular height from the base to each of the corners must therefore be the same, and thus the volume of each tetrahedron is the same.

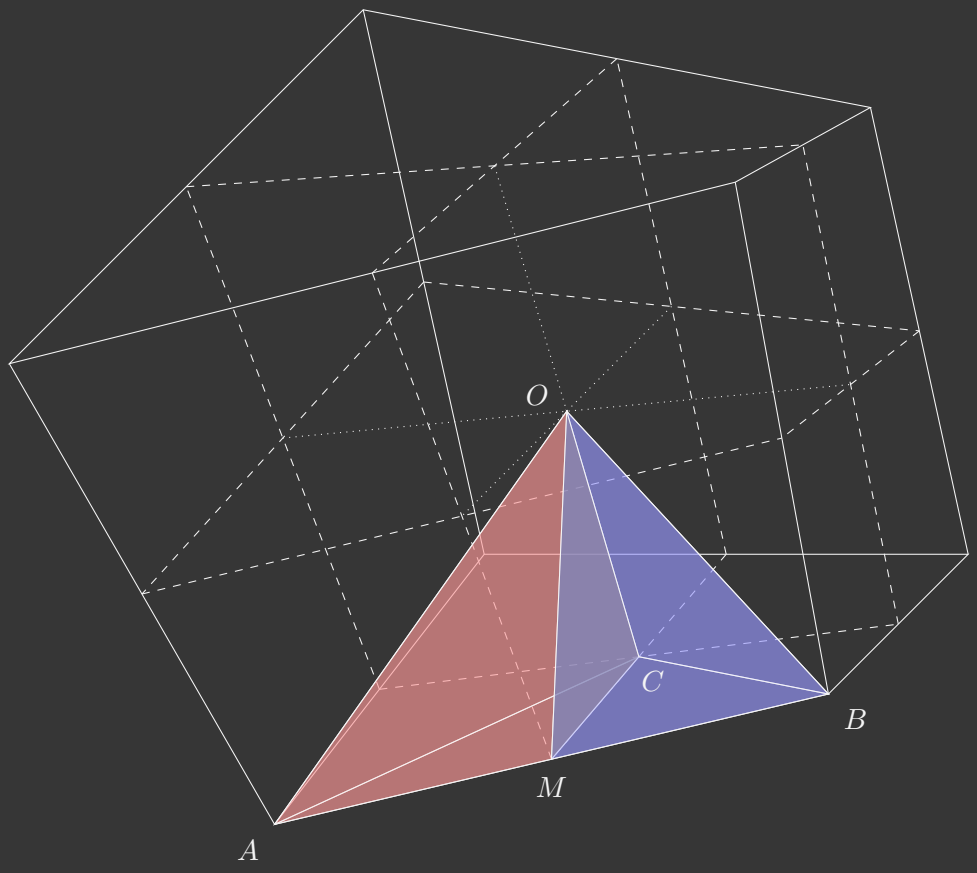


Figure 9: The $n = 2$ three-dimensional wonky chessboard can be decomposed into 24 pairs of tetrahedra of equal volume.

[5]This can be seen by considering a $2 \times 2 \times 2$ cube which can be thought of as a special case of the three-dimensional wonky chessboard that is non-wonky. If we have decomposed the non-wonky chessboard into tetrahedra then we see that moving the vertices around will maintain this decomposition (as long as each of the vertices are vertices of the convex hull of the vertices). For example, if two tetrahedra share a face then they must share the three vertices that define that face. Moving any of these vertices around will move where the face is, but the face must still be shared between the two tetrahedra

3. Given $n+1$ real numbers, $x_0, x_1, \ldots, x_{n-1}, x_n$, and a real-valued function f, there exists a polynomial, p, of degree at most n that interpolates f at these points. That is $y_i = f(x_i) = p(x_i)$ for all x_i, $i = 0, 1, \ldots, n-1, n$. This polynomial is unique and is known as the Lagrange Interpolating Polynomial, or simply the Lagrange Polynomial.

   (a) If f is a quartic polynomial satisfying $f(1) = 1$, $f(2) = 4$, $f(3) = 9$, $f(4) = 16$, and $f(5) = 35$, determine $f(6)$. Note that $f(5) \neq 5^2$.

   Hints:

   1. Consider the function $g(x) = f(x) - x^2$.
   2. Note that $g(x)$ is also a quartic polynomial and has roots at 1, 2, 3, 4.
   3. As $g(x)$ is a quartic, by the factor theorem it must be of the form $\lambda(x-1)(x-2)(x-3)(x-4)$ for some $\lambda \in \mathbb{R}$.

   Solution: Building on the hints, we only need to determine λ to completely determine $g(x)$ and thus $f(x)$. This can be done with simple algebra as we know that $g(5) = f(5) - 5^2 = 10$, but we can make our lives easier. Consider the function given in equation 4. When we substitute in $x = 5$, the whole fraction simplifies to 1 and gives $\mu = 10$. Note that we are free to do this as the denominator is simply a constant so $h(x)$ is in exactly the same form as $g(x)$. Determining $h(6)$ causes diagonally adjacent factors in the fraction to cancel and we get $h(6) = 10 \cdot \frac{6-1}{5-4} = 50$, therefore $f(6) = 6^2 + 50 = 86$.

$$h(x) = \mu \frac{(x-1)(x-2)(x-3)(x-4)}{(5-1)(5-2)(5-3)(5-4)} \tag{4}$$

   (b) f is a degree n polynomial satisfying $f(k) = k^2$ for $k = 0, \ldots, n-1$ and $f(n) = \alpha$. By using the ideas used in part a or otherwise, determine $f(x)$. You do not need to give your answer in the form $a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$.

   Solution: We follow the same procedure from a, but with more generality. The corresponding $h(x)$ can be written down immediately as in equation 5 and the final answer is given as $f(x) = h(x) + x^2$.

$$h(x) = \alpha \cdot \frac{\prod_{k=1}^{n}(x-k)}{\prod_{k=1}^{n}(n-k)} \tag{5}$$

   (c) By using the ideas you may have used in part b or otherwise, find an explicit formula for an interpolating polynomial. This polynomial does not necessarily need to be of minimum order.

   Hints:

   1. Use linearity - the finite sum of a collection of polynomials is also a polynomial.
   2. Find a way to focus on one point at a time.
   3. Use the factor theorem - a polyomial has a root at x_0 if and only if the polynomial is a multiple of $x - x_0$.

   Solution: The key to this problem is to use the fact that polynomials can be added to get another polynomial. This allows us to isolate each point in the following way. For each

point, (x_i, y_i), we can find a polynomial, $p_i(x)$, that is 0 when evaluated on all x_j where $j \neq i$ and satisfies $p(x_i) = y_i$. The sum of these polynomials gives us an interpolating polynomial by equation (6) ($\delta_{ij} = 0$ if and only if $i = j$).

$$p(x_i) = \sum_{j}^{n} p_j(x_i) = \sum_{j}^{n} y_i \delta_{ij} = y_i \tag{6}$$

The problem now reduces to finding each of the p_i. In fact we can simplify the problem further as we only need to find a function, q_i, that is 0 for x_j for $j \neq i$ and satisfies $q_i(x_i) \neq 0$. Once such a function has been found, we can define p_i as in equation (7a). The factor theorem can then be used to deduce q_i as follows: x_0 is a root of p if and only if $x - x_0$ divides $p(x)$. This gives q_i as the product in equation (7b). The final polynomial is given in equation (7c) by combining these parts.

$$p_i(x) = \frac{q_i(x)}{q_i(x_i)} \cdot y_i \tag{7a}$$

$$q_i(x) = \prod_{\substack{j=1 \\ j \neq i}}^{n} (x - x_j) \tag{7b}$$

$$p(x) = \sum_{i=1}^{n} \prod_{\substack{j=1 \\ j \neq i}}^{n} \frac{x - x_i}{x_j - x_i} y_i \tag{7c}$$

(d) Show that there is only one interpolating polynomial of minimal order. That is, if for all x_i, $i = 0, 1, \ldots n-1, n$ we have $p(x_i) = f(x_i) = q(x_i)$ then $p(x) = q(x)$ for all $x \in \mathbb{R}$. Deduce that all polynomials of degree n or less that interpolate $n + 1$ points must be of minimum order.

Hints:

- Consider $p(x) - q(x)$.
- What are the roots of $p(x) - q(x)$?

Solution:

As both p and q are interpolating polynomials, $p(x_i) - q(x_i) = 0$ for all x_i. As $p(x) - q(x)$ is of degree n and has roots at $n + 1$ points, it must be identically 0. This can be seen by the factor theorem as $p(x) - q(x) = \prod_{i=0}^{n} \lambda(x - x_i)$ for some $\lambda \in \mathbb{R}$, yet for $\lambda \neq 0$ the right hand side has degree $n + 1$. We conclude that $\lambda = 0$ and that $p(x) = q(x)$.

If there is some polynomial, p, that interpolates f at $n+1$ points then it necessarily agrees with the interpolating polynomial of minimal degree at these points. By the previous argument we must have that p is identically equal to the polynomial of minimal degree.

4. (a) Which is bigger: $\sqrt{2}$ or $\sqrt[3]{3}$?

Solution:

The radicals can be eliminated by raising both sides to the power of 6. Which number is bigger is preserved by this because both are positive and $f(x) = x^6$ is an increasing function. The comparison turns into which is bigger of $2^3 = 8$ and $3^2 = 9$, we therefore conclude that $\sqrt[3]{3} > \sqrt{2}$.

(b) Estimate $\sqrt{14}^8$

Solution:

Trying to simplify $\sqrt{14}$ by approximating it as $\sqrt{16}$ gives a very crude approximation. Equation (8) gives two more precise ways to proceed.

$$\sqrt{14}^8 = 14^4 = 196^2 \approx 200^2 = 40000 \tag{8a}$$

$$\sqrt{14}^8 = 14^4 = 2^4 \cdot 7^4 = 16 \cdot 49 \cdot 49 \approx 16 \cdot 50 \cdot 50 = 40000 \tag{8b}$$

If we know that $49^2 = 2400$ then we can get a slightly more complicated multiplication, but gives a very good approximation of $16 \cdot 2400 = 38400$. This is essentially as much work as computing the exact answer which is 38416.

5. How might you design a dam?

Hints:

1. What materials should be used?
2. What shape should the dam be? Describe the cross-section.

Solution: This question was deliberately open-ended so interviewees were expected to come up with what aspects would be important to consider. The following are suggested.

- Porosity. The material would need to be non-porous so that water does not leak through. This rules out cheap materials such as sand and earth.
- Corrosion. Water can react with many materials which can increase maintenance requirements or compromise the dam integrity.
- Cost. Dams are very large so making them out of exotic materials or with an intricate design would be expensive. Even materials such as steel would be prohibitively expensive given the quantities that are needed.
- Strength. The dam will need to resist a lot of force.

Candidates are anticipated to arrive at a giant uniform mass of concrete or other suitable material. It remains to determine what shape the dam should be so that materials are used efficiently. Water has a high density which means the pressure that a dam needs to hold back can be very large - understanding this pressure is what determines the shape of the dam.

Stevin's law states that the rate of change of pressure in the vertical direction within a static fluid is directly proportional to the specific weight of the fluid. As water can be approximated as incompressible we can assume a constant density, therefore the hydrostatic pressure increases linearly with depth of water. Pascal's law states that the pressure in a fluid acts equally in all directions, therefore the pressure would also act against the walls of the dam.

For the dam to withstand the higher pressure it would need to be stronger towards the base. We can assume that the strength of a wall (that is, how much pressure it can withstand) is proportional to its thickness, therefore the dam wall thickness will increase linearly with the depth of the water.

6. How many equilateral triangles can be formed by connecting three vertices of a cube?

Hints:

1. Explore the problem. For example, if one of the edges of the cube is included in the triangle, what are the possible triangles you get by adding one of the other vertices?
2. If a triangle shares an edge with the cube, it must be a right-angled triangle.
3. Equilateral triangles have rotational symmetry of order 3 - where on a cube do you find such a symmetry?

Solution:

Place three points overlapping at a vertex of the cube. There are three edges connected to this vertex, and we move the points along each of the edges. If all points move along their edge at the same rate then they will form an equilateral triangle by symmetry. Eventually they hit the vertex at the end of the edge, and thus we have found three vertices that form an equilateral triangle. We note that each triangle generated this way cannot be reached by starting at any other vertex, thus we have one triangle per vertex and the final answer is at least 8. This process is demonstrated in figure 10. The red triangle shows the intermediate step where the points are partially along the edges and the blue triangle shows where it stops.

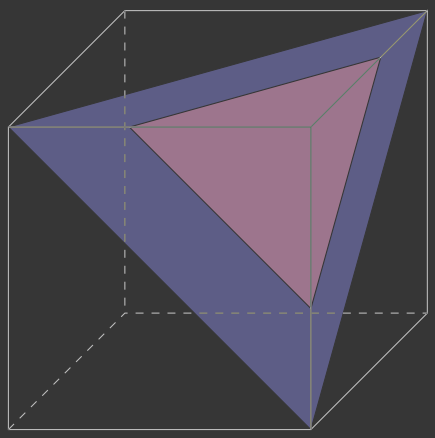


Figure 10: Three points moving away from a vertex at the same rate form an equilateral triangle by symmetry.

Another way of producing this lower bound is by considering symmetry. There are 24 orientation-preserving symmetries of the vertices of a cube which form the O group. The vertices of an equilateral triangle have 3 orientation-preserving symmetries and form the C_3 group. The quotient set O/C_3 has size $|O/C_3| = \frac{|O|}{|C_3|} = \frac{24}{3} = 8$ which shows there are at least eight triples of vertices with threefold rotational symmetry, in other words, the equilateral triangles. This is only a lower bound because we could have shapes such as two cubes next to each other that share a face. This shape has 8 symmetries which is not a multiple of three, yet clearly it has at least 16 equilateral triangles in it.

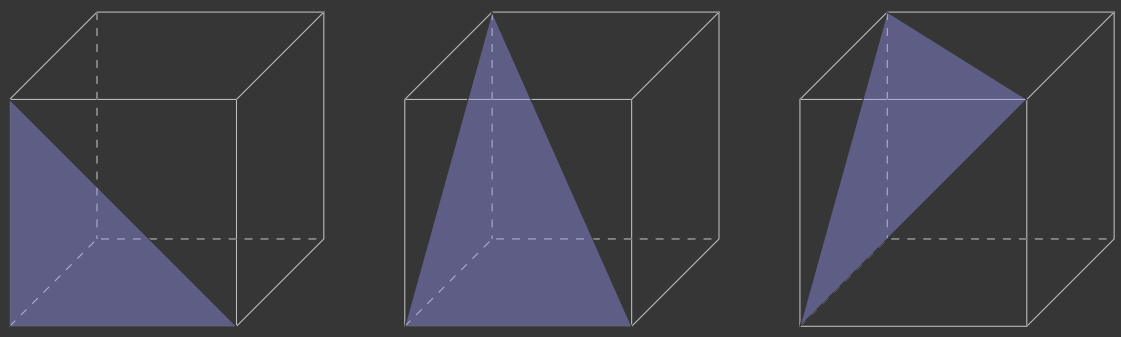
We can confirm that the triangles associated with each vertex are the only such equilateral triangles by classifying all triangles within the cube. If we have lower bounds for all types of triangle which sum to the total number of triangles then our bounds must be tight. We will find it helpful to first define the distance between two points to be the minimum number of edges of the cube needed to be traversed to navigate from one point to the other.

Consider a triangle that shares an edge with the cube - we will call this the base edge. If the remaining vertex is within a distance of one from the base edge the three edges will lie on the same face. There are four triangles per face and six faces of the cube so there are 24 such triangles. The furthest a vertex can be from the base edge is a distance of two, occuring when it lives on the edge opposite the base edge. There are twelve choices for the base edge and two possible vertices a distance of two away, giving 24 such cases. All 48 of these cases are right-angled, therefore we are done as there are 56 possible triangles.

Even though we have already identified all the triangles, for the sake of interest we see we could have found them by continuing our classification. We cannot have edges within a distance of 1 of each other or we will find ourselves in the first two cases considered. The furthest apart two vertices can be is a distance of three, but then anywhere we put the third vertex must be within a distance of one of an existing vertex. Having two vertices a distance of three apart is in fact the same as our second case. The only remaining case is where all vertices are a distance of two apart from each other which are our equilateral triangles. All three cases are summarised in 1 and shown in figure 11.

Table 1: Enumerating all possible triangles formed by connecting vertices of a cube.

Classification	Number of triangles
Contained within a face	$4 \cdot 6 = 24$
Only one edge within a face	$12 \cdot 2 = 24$
All edges are internal	8
Total	$\binom{8}{3} = 56$



(a) Contained within a face (b) Only one edge within a face (c) All edges are internal

Figure 11: Examples of the three distinct ways a triangle can be inscribed within a cube.

7. There is an infinite grid of non-negative integers where each number is equal to the mean of its four directly adjacent neighbours. Characterise all such solutions with proof.

Hints:

1. The only solutions are the constant solutions, that is, every number is the same.
2. In one dimension the mean of neighbours property forces all numbers to be in an arithmetic sequence.
3. Prove that the only solutions are constant by contradiction - assume that a non-constant solution exists and show that this breaks the rules of the grid.
4. If non-negative integers were allowed, there would exist non-constant solutions. From this you know that the non-negativity must be an important part of your proof.

Solution:

We note that adding a constant to the whole grid will not change whether something is a solution. If we were to find a non-constant solution where the minimum was -10 for example, we could add 10 to all numbers in the grid and we would have a non-negative constant solution. The non-negativity is simply providing a lower bound, but the fact that this lower bound is at 0 is arbitrary.

Playing around with random numbers and trying to build the grid, we note that when one neighbour is bigger, it seems that we need another to be smaller to compensate for it. This makes us think that in a non-constant solution there may be some strictly decreasing sequence of numbers which would break our lower bound and give us a contradiction.

If the solution is non-constant and bounded from below then there exists some minimum number, m, with neighbours a, b, c, and d where one of them is different to m. Without loss of generality we choose $a \neq m$, and by minimality of m we have $a > m$. Equation (9) gives us our contradiction using the mean-property. Explicitly, m cannot be the minimum of the whole grid because one of its neighbours would have to be even smaller than m.

$$0 = a + b + c + d - 4m \geq a + m + m + m - 4m = a - m > 0 \tag{9}$$

For interest only we note that this result is also true for non-negative real numbers, but the above proof does not work. By the well-ordering property we can deduce the existence of a minimum over the grid of integers, but there is not necessarily a minimum over a grid of real numbers. For example, the sequence $a_n = \frac{1}{n}$ for $n \in \mathbb{N} \setminus \{0\}$ is not bounded below by any number greater than 0. The minimum must be attained, and $\frac{1}{n} \neq 0$ for all n so the sequence has no minimum[6].

[6]The greatest upper bound and the smallest lower bound are called the infinum and the supremum respectively, and these always exist for a set of real numbers.

8. The integers from 1 to n^2 are put into an $n \times n$ grid.

   (a) Prove that there must exist one pair of numbers that are adjacent (either horizontally, vertically, or diagonally) that are within $n+1$ of each other.

Hints:

1. Use the pigeonhole principle.
2. The adjacency part of the question motivates us to think about 3×3 regions. This ends up not being very useful as such regions give control over too small a region. Try and get control in another way.
3. Think about paths through the grid.

Solution:

We first build some intuition by considering the first non-trivial case, $n = 3$. All squares other than the central square must see the central square. If the centre square as less then 5 then the difference to 9 would be greater than 4 and if the centre square is greater than 5 then the difference to 1 would be greater than 4. Therefore 5 is the best we can do and this has a gap of 4, exactly $n+1$. Changing the value of the centre square to reduce a gap on one side causes a larger gap on the other side. This motivates thinking about paths through the grid.

There are many different paths through the grid so we need a way of gaining control over them. Define a step to be a move to an adjacent square (either horizontall, vertically, or diagonally) and consider moving from one square to another and trying to minimise the number of steps taken. There are many choices that give the same number of steps and we see that as long as each step moves nearer to the second square in the direction that the two squares are greatest apart in, the number of steps will be the same. This induces a notion of distance between two squares given by the maximum of the number of horizontal steps and the number of vertical steps between the squares.

n^2 and 1 must be connected by a path of squares at most $n-1$ steps long. If each step was between squares that were at most n apart then the largest difference that could be achieved in $n-1$ steps would be $n^2 - n < n^2 - 1$ for $n > 1$. By the pigeonhole principle, we have proved there must exist one step between squares with a difference of $n+1$ or higher. In fact $(n-1)(n+1) = n^2 - 1$ so if the smallest difference between adjacent squares was $n+1$ then there exists a path between 1 and n^2 where all differences between squares on the path are equal to $n+1$.

(b) Is this bound tight? That is, does there necessarily exist a pair of numbers that are $n+2$ apart? Either prove the result from part a for $n+2$ or find a counter example.

Hints:

1. The bound is tight.
2. To construct a counter example, we want numbers to be close together. Write down the $n = 5$ case for example of the simplest way of filling in the grid that might work and check whether it is a counter example. How can this be generalised?

Solution:

The result is not true for $n+2$. As a counter example, we fill in the $n \times n$ grid with numbers sequentially, going along the rows first. We look at column x and row y and consider the adjacent points. If the number lies on an edge then we consider only the adjacent points that exist. The number at this point has value $ny + x$ and we see that the differences to the adjacent points are bounded by $n + 1$, as shown explicitly in table 2.

Table 2: The difference in value between the number at (x, y) and its adjacent neighbours.

Direction	Coordinates	Value	Difference
Top Left	$(x-1, y-1)$	$ny - n + x - 1$	$n+1$
Top Centre	$(x, y-1)$	$ny - n + x$	n
Top Right	$(x+1, y-1)$	$ny - n + x + 1$	$n-1$
Left	$(x-1, y)$	$ny + x - 1$	1
Right	$(x+1, y)$	$ny + x + 1$	1
Bottom Left	$(x-1, y+1)$	$ny + n + x - 1$	$n-1$
Bottom Centre	$(x, y+1)$	$ny + n + x$	n
Bottom Right	$(x+1, y+1)$	$ny + n + x + 1$	$n+1$

9. There are 25 horses of unknown speed. They can be raced in groups of five to find the finishing order, but their times are not recorded. What is the minimum number of races necessary to determine the fastest three horses and what strategy should be used to accomplish this?

Hints:

1. Can you think of an easier problem involving the 25 horses?
2. How would you determine the fastest horse?
3. The second-fastest horse may have been in the same race as the fastest horse. Keep track of where the possible fast horses may be and then race them together.
4. Seven races are necessary and sufficient.

Solution:

All horses need to be raced at some point and 25 naturally decomposes into five races of five horses which we will call the heats. The fastest horse must be among the winners of the heats and racing these five together in a sixth race determines the winner. We label the heats A, B, C, D, and E in order of how fast the fastest horse in each race was and we label the horses within each race by their position. For example, B3 is the horse that came third in the heat with the horse that came second in the sixth race.

The second fastest horse would have either won their heat or have been beaten by the fastest horse if they shared the same heat. This means the second fastest horse was either A2 or B1. We cannot distinguish between these cases with the information available and there is clearly no other way of finding the fastest horse in six races. We therefore conclude that a seventh race is necessary. The third fastest horse could be any of the following three:

1. A3: This is where the second fastest horse was in the same heat as the fastest horse and both of them beat the third fastest.
2. B2: This is where the second fastest horse was in a different heat from the fastest horse, but shared a heat with the third fastest.
3. C1: The top three fastest horses were in different heats so all made it through to the sixth race.

Together this makes A2, B1, A3, B1, and C1 which is five horses. If these horses are raced together then the top two positions give the second and third fastest horses. The different possibilities for the final race are given in table 3.

Table 3: Final race possibilities based on the initial position of the fastest three horses.

Race 1	Race 2	Race 3	Second Horse	Third Horse
1, 2, 3			A2	A3
1, 2	3		A2	B1
1	2, 3		B1	B2
1, 3	2		B1	A2
1	2	3	B1	C1

10. Sketch $f(x) = \sin \frac{1}{x^2}$.

Solution: Before we start thinking about what shape the function is, we make a comment about its domain. sin is defined on all real numbers (and even complex numbers), but $\frac{1}{x^2}$ is not defined at $x = 0$. We conclude that f is defined for all non-zero real numbers and that there may be some strange behaviour around 0.

We can simplify the problem by noting symmetries. We see that sin is an odd function and $\frac{1}{x^2}$ is an even function. The composition of an even function with an odd function is an even function so we expect the graph to be symmetrical about the y-axis.

There are two natural limits to consider, $x \to \pm\infty$ and, as noted earlier, $x \to 0$. As $x \to \pm\infty$, $\frac{1}{x^2} \to 0$ and $\sin 0 = 0$, therefore we expect f to converge to 0 in both directions. Strong candidates may also note that we have moved the limit inside the function so we need to be careful, but this is valid because sin is continuous. As $x \to 0$, $\frac{1}{x^2} \to \infty$, and $\sin t$ oscillates as t increases, therefore we expect oscillatory behaviour at 0. The function must oscillate between -1 and 1 so the function cannot be extended to a value at 0.

sin has a fixed period but $\frac{1}{x^2}$ increases faster than linearly as $x \to 0$ so we expect the oscillations to get faster. The period of sin is 2π, and $\frac{1}{x_n^2} = 2n\pi \, n \in \mathbb{N} \implies x_n = \frac{1}{\sqrt{2n\pi}}$. We can compute the gap between periods, $x_{n+1} - x_n$, explicitly and using $\sqrt{1 + 1/n} = 1 + \frac{1}{2n} + \mathcal{O}(\frac{1}{n^2})$ we see that size of the gaps are $n^{-\frac{3}{2}}$ to leading order.

Finally we determine the sign of $f(x)$ as $x \to \pm\infty$. As $\frac{1}{x^2}$ is small and positive and $\sin(t) > 0$ for small values of t we see that $f(x)$ is also positive for large enough $|x|$. A plot of f is shown below in figure 12.

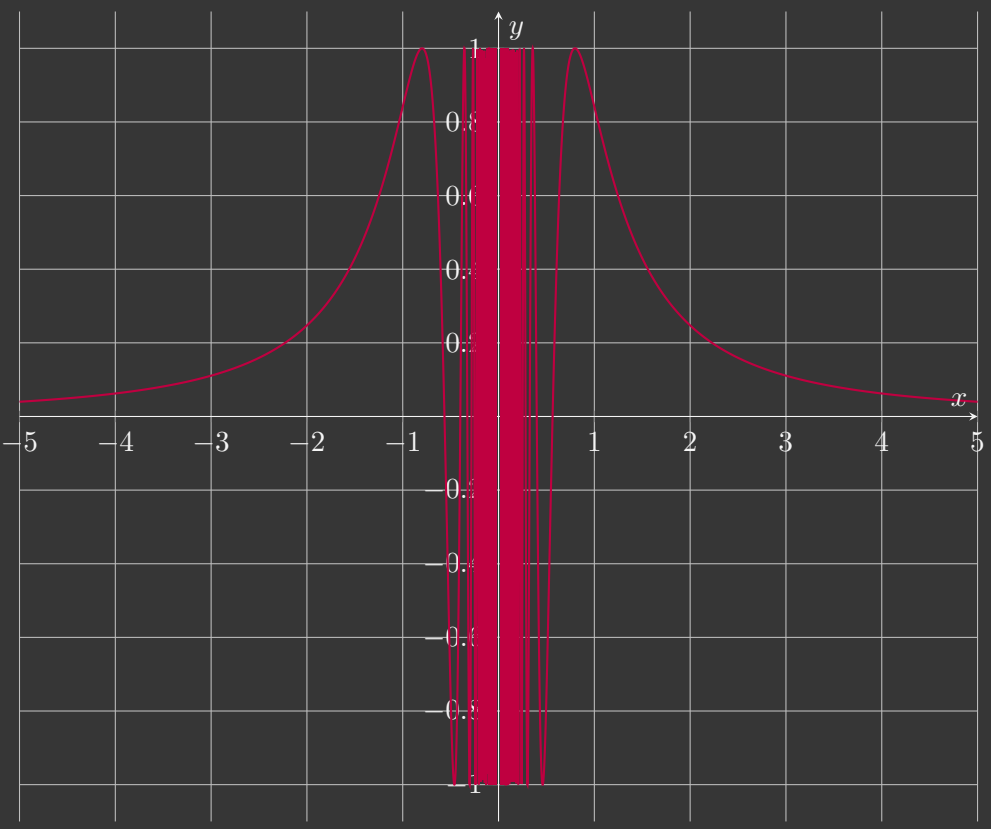
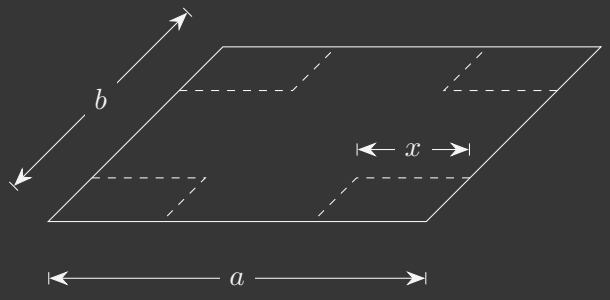
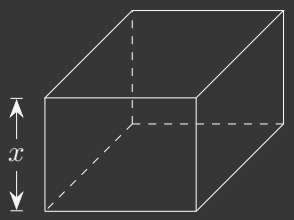


Figure 12: $y = \sin \frac{1}{x^2}$

11. Consider a rectangle with side lengths a and b. Squares with a side length of x are cut out of each corner as shown in figure 13a. A box without a top is then formed by folding the sides up, as shown in figure 13b. What value of x maximises the volume of the box for a given a and b?



(a) Cuts made into the rectangle



(b) Box after folding

Figure 13: Creating a box from a rectangle

Hints:

1. Form an equation for the volume, differentiate, set equal to 0 and solve.
2. You also need to prove that the value you get is a minimum.

Solution:

The height of the box is x and the side lengths are $a-2x$ and $b-2x$. This gives the volume of the box as $f(x) = x(a-2x)(b-2x) = 4x^3 - 2x^2(a+b) + abx$. Differentiating gives $f'(x) = 12x^2 - 4x(a+b) + ab$ which has roots at $x_\pm = \frac{1}{6}\left(a+b \pm \sqrt{a^2-ab+b^2}\right)$.

We present two ways to show that this is a minimum. The first method uses the second derivative as given in equation 10. This shows that x_- is a maximum as desired as $f''(x_0) < 0$ and x_+ can be rejected.

$$f''(x_\pm) = 24x_\pm - 4(a+b) = 4\left(a+b \pm \sqrt{a^2-ab+b^2} - (a+b)\right) = \pm\sqrt{a^2-ab+b^2} \tag{10}$$

The second method uses the fact that geometrically we only expect a single stationary point. At $x = 0$ and $x = \min\frac{a}{2}, \frac{b}{2}$ the volume is 0, therefore this stationary point should be a maximum. It is therefore sufficient to reject one of the solutions we found and we do this by showing that $x_+ > \min\frac{a}{2}, \frac{b}{2}$. Without loss of generality we assume $a < b$ and assume for a contradiction that $x_+ < \frac{a}{2}$. This is done in equation 11.

$$\begin{aligned} & x_+ < \frac{a}{2} \\ \implies & \frac{1}{6}\left(a+b+\sqrt{a^2-ab+b^2}\right) < \frac{a}{2} \\ \implies & a+b+\sqrt{a^2-ab+b^2} < 3a \\ \implies & a^2-ab+b^2 < (2a-b)^2 = 4a^2-4ab+b^2 \\ \implies & 0 < 3a^2-3ab = 3a(a-b) < 0 \end{aligned} \tag{11}$$

12. There is a 1 L jug filled 90% of water by volume with the remaining 10% filled with alcohol. There is a hole at the bottom of the jug which causes 100 ml/s of the liquid to come out. Water enters the jug at 100 ml/s.

    (a) Assuming perfect mixing of water and alcohol, come up with an equation for the concentration of alcohol in the jug over time and solve it.

    Hints:

    1. Make an equation for the volume of alcohol in the jug.
    2. The volume of fluid in the jug is constant.
    3. Find an equation for the alcohol leaving the jug.
    4. If you have not covered first order constant coefficient ordinary differential equations, note that $\frac{\mathrm{d}}{\mathrm{d}x}e^{mx} = me^{mx}$.

    Solution:

    As the volume entering the jug is equal to the volume leaving, the total volume remains constant. We define the volume of alcohol in litres of alcohol per litre of fluid in the jug to be x. If the fluid leaves at r l/s then rx of this is alcohol. The amount of alcohol entering the jug is 0 l/s so the overall rate of change of alcohol in the jug by volume is $\frac{\mathrm{d}x}{\mathrm{d}t} = -rx$. Solving this we get $x = r_0e^{-rt}$ where r_0 is the initial volume of alcohol in the jug initially. Substituting $r = 0.1 = r_0$ applies our boundary conditions and the volume of alcohol is given as $0.1e^{-0.1t}$.

    (b) Discuss the validity of the assumption that the alcohol mixes perfectly with the water.

    Hints:

    1. Will the true concentration be higher or lower than the model predicts?
    2. Consider the case with no mixing.
    3. Draw a graph to compare the two.

    Solution:

    A proper assessment of mixing would need to account for the geometry of the jug, the nature of the flow of water entering the jug, and the turbulence of the fluids inside the jug. It is often not possible to explore all assumptions in this level of detail so we need to apply judgement in our modelling. Given that the entire volume of fluid that the jug holds is replaced every ten seconds, we can assume that the water is reasonably turbulent and therefore should have a good level of mixing. If the mixing was poorer than expected then the concentration of alcohol would likely be higher at the bottom of the jug, therefore out prediction would be an overestimate.

    We explore this by considering the other extreme case where this is no mixing at all as the true answer will likely lie in between these cases. To determine the concentration of alcohol in the jug without mixing we can think of a second jug without any water entering it. The concentration of alcohol in this jug will remain constant, so $\frac{\mathrm{d}x}{\mathrm{d}t} = -rx_0$ where x_0 is the number of litres of alcohol originally (and therefore equal to the concentration). When this second jug empties the rate of alcohol leaving will be 0. Integrating gives equation (12),

and as before the concentration is equal to the number of litres of alcohol because the total volume is a constant 1 L.

$$\frac{\mathrm{d}x}{\mathrm{d}t} = \begin{cases} -rx_0 & \text{if } 0 \leq t < \dfrac{1}{r} \\ 0 & \text{if } \dfrac{1}{r} \leq t \end{cases} \quad \Longrightarrow \quad x = \begin{cases} x_0(1-rt) & \text{if } 0 \leq t < \dfrac{1}{r} \\ 0 & \text{if } \dfrac{1}{r} \leq t \end{cases} \tag{12}$$

This is plotted in figure 14. We see that the gradients match at $t = 0$ and the two solutions are close for small values of t, implying that our assumption is good in this region. For larger values of t the solutions stray apart, reaching a maximum difference at $t = \frac{1}{r}$. The no-mixing solution is expected to be a poor representation of reality in this region which is also indicated by the unphysical discontinuous first derivative so this difference is perhaps expected. As $t \to \infty$ the solutions converge but this method is too crude to say anything more about the accuracy of this convergence rate.

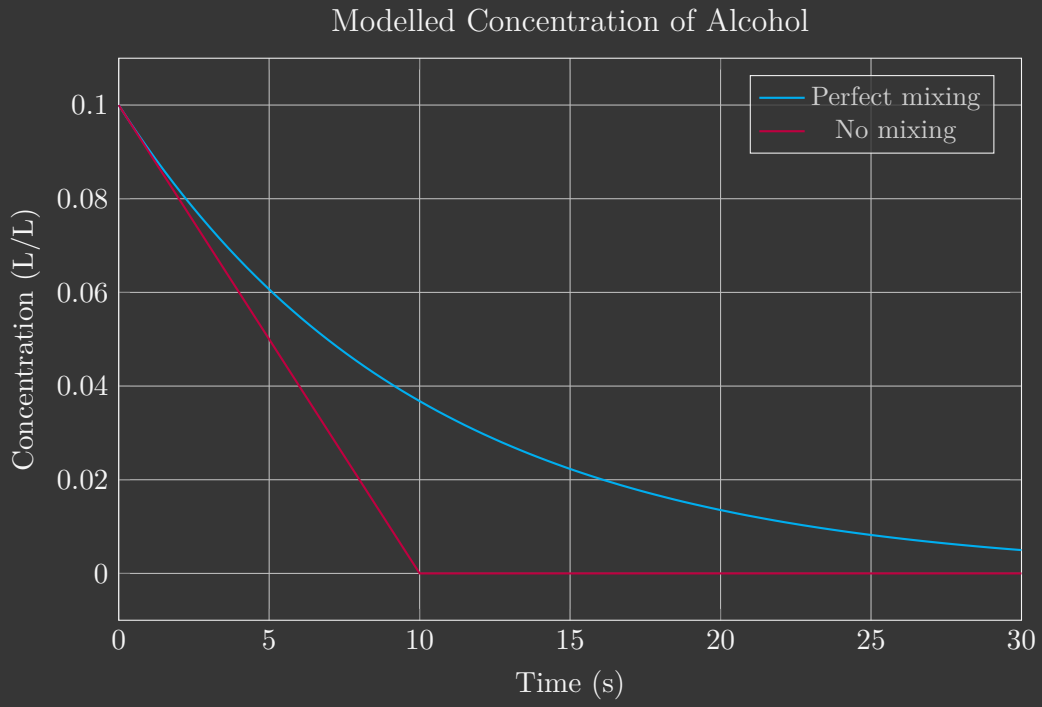


Figure 14: Comparing the case of perfect mixing with no mixing.

We conclude that our initial assumption is likely quite good, although perhaps the decay constant may be slightly higher than our model predicts. This last note would be especially important to consider for tall and thin jugs.

13. Find the centre of mass of a plane

Solution:

FOR ROB TO FILL IN.

14. (a) Start with the numbers 1, 2, 3, 4, 5, 6, 7. Remove any two numbers, a and b, from the list and replace them with $ab + a + b$. Repeat this until only a single number remains. What is the final number?

Hints:

1. Attempting this naively by hand will be computationally hard - do not reach for your calculator, change your approach.
2. Do the process algebraically for a, b, and c - what do you notice?.
3. Expand $(x - a)(x - b)(x - c)$

Solution:

When doing this with a list of three numbers, a, b, and c, we get $abc + ab + bc + ca + a + b + c$. This looks very similar to the coefficients in an expansion of the product of linear terms and we see this to be the case in equation (13). This process can be seen as equivalent to expanding these brackets, where choosing two numbers to replace is the same as choosing the corresponding pair of brackets. The answer follows from commutativity and associativity of real numbers. The final answer is $8! - 1$.

$$abc + ab + bc + ca + a + b + c = (1 + a)(1 + b)(1 + c) - 1 \tag{13}$$

(b) Prove the formula you arrived at in the part a for any starting numbers and show that the result is independent of the choices made. What if we replaced a pair of numbers, a and b, with $\lambda ab + \mu(a + b)$ where $\lambda \geq 0$?

Hints:

1. The formula is $-1 + \prod_{i=1}^{n}(1 + a_i)$ where a_i are the starting numbers.
2. Use induction on the length of the initial sequence.

Solution:

This step is essentially just formalising the ideas used in the first part. We can define A_n to be a tuple of numbers with $|A_n| = n$. We denote the function that maps A_n to the result of the merging process as f and do induction on n to prove that $f(A_n) = -1+\prod_{i=1}^{n}(1+a_i)$ where $A_n = (a_1, \ldots, a_n)$. The base case where $n = 1$ is trivial as $f(A_1) = (1+a_1)-1 = a_1$. The induction step follows the same reasoning from before and is given in equation (14).

$$\begin{aligned} f(A_n) &= f(A_{n-1}) \cdot a_n + f(A_{n-1}) + a_n \\ &= -1 + (f(A_{n-1}) + 1)(a_n + 1) \\ &= -1 + \left(\prod_{i=1}^{n-1}(1 + a_i)\right)(a_n + 1) \\ &= -1 + \prod_{i=1}^{n}(1 + a_i) \end{aligned} \tag{14}$$

For the last part we note that $(\alpha a+\beta)(\alpha b+\beta) = \alpha^2 ab+\alpha\beta(a+b)+\beta^2$. Defining $\alpha = \sqrt{\lambda}$ and $\beta = \mu$ matches the coefficients in the replacement formula, therefore applying the above argument shows that the generalised formula is $-(\mu^n) + \prod_{i=1}^{n}(\sqrt{\lambda}a_i + \beta)$.

15. A ball is in a boat floating in a pool of water, as shown in figure 15. In the following scenarios, does the water level go up, go down, or stay the same?

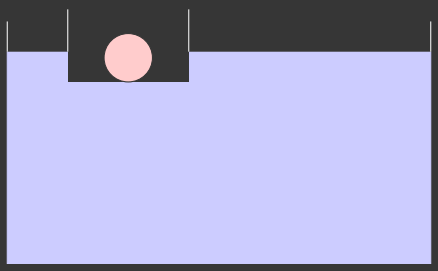


Figure 15: A boat with a ball in it floating in a pool.

(a) The ball is thrown into the water and floats.

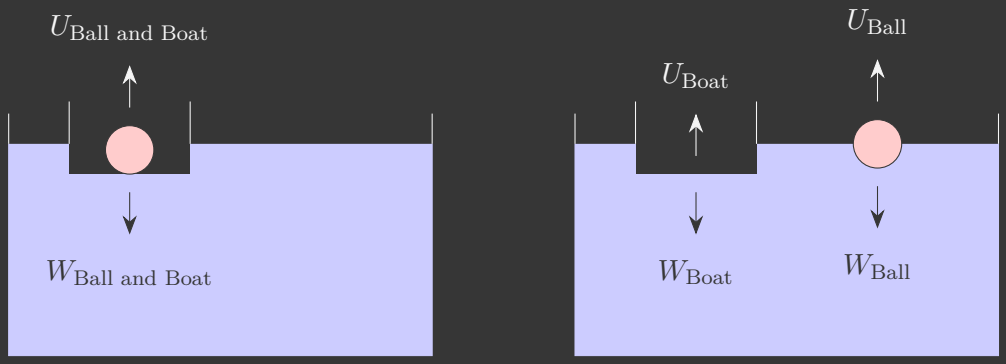
Hints:

1. Draw a force diagram for before and after the ball is put in the water.
2. Archimedes' principle states that the weight of water displaced by an object is equal to the buoyancy force (upthrust) acting upwards on that object.

Solution:

The key to this question is understanding the forces acting, in particular whether the buoyancy force acts due to the weight of the ball or the volume of the ball. When the ball is in the boat the water level is a function only of the ball's weight and is independent of the ball's volume. When the ball is floating in the water there are only two forces acting on it - the upthrust from the buoyancy of the ball and the weight, as shown in figure 16b.

The ball is in equilibrium so by Newton's first law the forces acting on the ball must be in balance, therefore $U_{\text{Ball}} = W_{\text{Ball}}$. By Archimedes' principle, the upthrust acting on the ball is also equal to the weight of the water displaced by the ball. The total weight of water displaced is therefore equal to the weight of the boat and the weight of the ball, exactly the same weight that was displaced when the ball in the boat, as shown in figure 16a.



(a) Forces on the ball and boat system

(b) The ball floating on the surface

Figure 16: Comparing the forces on the ball and boat system to the separated ball and boat systems.

As the density of water is the same before and after the ball is removed from the boat, the volume of water displaced is proportional to the mass of water displaced. As the strength of gravity is also constant, the mass of water displaced is proportional to the weight of

water displaced. The water level is a strictly increasing function of the volume of water displaced, and putting all these relationships together we deduce that the water level stays the same.

(b) The ball is thrown into the pool and sinks.

Hints:

1. An object sinks if the upthrust is less than its weight. How does the system remain in equilibrium?
2. Compare forces before and after the ball is removed from the boat.

Solution:

In this situation the ball is sufficiently dense where it cannot displace enough water to produce a large enough buoyancy force to support its own weight. The bottom of the pool therefore needs to provide a reaction force acting upwards on the ball to keep the ball in equilibrium, as shown in figure 17.

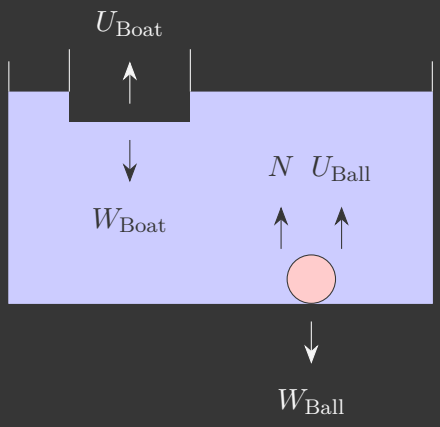


Figure 17: The ball resting on the bottom of the pool.

The total downwards force acting on the ball and the boat combined is the same as when the ball was in the boat, $W_{\text{Ball}} + W_{\text{Boat}}$. As the total upwards force includes a strictly positive normal force, N, we must have $W_{\text{Ball}} + W_{\text{Boat}} > U_{\text{Ball}} + U_{\text{Boat}}$. By the logic used to relate buoyancy forces to the water level from part a we conclude that the water level must decrease.

Another approach is to consider an extremely dense ball. When in the boat, the ball would cause a large volume of water to be displaced due to having a large weight, but when outside the boat the ball would displace very little due to its low volume. Comparing these situations means we can conclude that the water level decreases.

(c) The ball is thrown into the pool and floats, but is then tied to the bottom of the pool so it is completely submerged.

Solution:

This is the opposite situation to part b where we have a tension acting downwards instead of a normal force acting upwards. The forces are shown in figure 18 and a very similar argument shows that the water level increases. The analogue of the extremely dense ball argument instead uses an extremely low-density ball with similar reasoning.

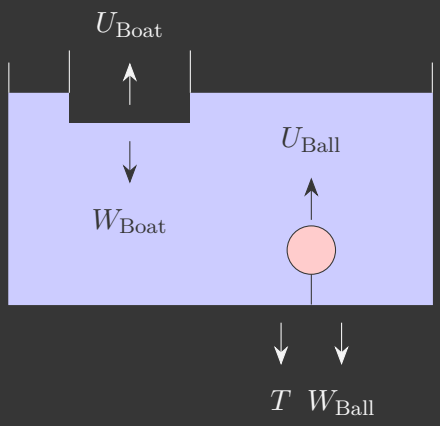


Figure 18: The ball is attached to the bottom of the pool.

(d) The ball is thrown into the pool and floats, but someone sitting in the boat pushes the ball underwater so that it is completely submerged.

Solution:

This case has almost identical logic to part a. The force diagram is shown in figure 19 where the only difference is the inclusion of a force, F. This force acts upwards on the boat and, by Newton's third law, has an equal and opposite force that acts downwards on the ball. Comparing the total upwards and downwards forces as in parts b and c, this force cancels out and gives the same equation that we would have found in part a. We conclude that the water level stays the same.

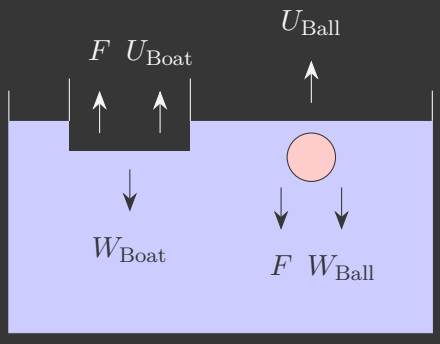


Figure 19: The ball is fully submerged because it is pushed under the water.

(e) The water in the pool is saltwater and the ball is made of ice (the ball can be assumed to be purely frozen freshwater). What happens when the ball melts?

Hints:

1. There are three densities involved here - ρ_{ice}, $\rho_{\text{freshwater}}$, and $\rho_{\text{saltwater}}$ - be careful to be precise which one you are referring to.
2. In the previous parts the ball was a rigid body but that is no longer the case once the ball has melted - how does this change things?
3. Even though saltwater is denser than freshwater, the melted ball cannot float above the surface because it is a fluid.

Solution:

We can simplify the problem by realising that the relative density of ice to that of saltwater or freshwater does not matter. The density of ice only effects the volume of the ball when it is in the boat which does not have any influence on the level of the water.

The volume of water displaced is equal to $(m_{\text{Ball}} + m_{\text{Boat}}) \cdot \rho_{\text{saltwater}} \cdot g$ when the ball is in the boat as before. When the ball is melted in the pool, the contribution of the volume of water displaced due to the boat is the same as before, $m_{\text{Boat}} \cdot \rho_{\text{saltwater}} \cdot g$, but the contribution from the ball is different. The melted water displaces $m_{\text{Ball}} \cdot \rho_{\text{freshwater}} \cdot g$ as it is completely submerged within the saltwater. As $\rho_{\text{saltwater}} > \rho_{\text{freshwater}}$, a larger volume of water is displaced and thus the water level rises.

16. There is an infinite square grid of lattice points. There are three pegs on three corners of a square as shown in figure 20a. A move consists of moving a peg over a peg to the opposite side, where the distance between the two pegs remains the same, as shown in figure 20b.



(a) Problem setup (b) Example move

Figure 20: The three pegs on a square problem

(a) Prove that the area of the triangle defined by the pegs is always $\frac{1}{2}$.

Solution:

1. There are many ways to find the area of a triangle, but the most useful formula to apply directly in this case is half the base times the height.
2. Show that for an arbitrary move, the area before and after the move is preserved.
3. Which edge of the triangle should be considered as the base?

If peg i is jumped over peg j, we define the base of the triangle to be the vector from peg i to peg j. The altitude of the triangle is the perpendicular height to the third peg from the line that passes through pegs i and j. After peg i has been moved, the altitude remains the same because peg i and peg j define the same line as before. The length of the base is also the same by definition of a move, and therefore the area is preserved.

(b) Is it possible to move one of the pegs to the last corner of the square? Prove it is not possible or find the fewest moves to do so.

Solution:

1. Label the pegs A, B, and C. Label the points that a peg lands on with the name of the peg that landed there.
2. Prove that the labelling in the previous hint is well-defined. That is, each square can only be landed on by a single type of peg. By labelling the fourth corner of the square as type D, deduce the conclusion.
3. It is not possible.
4. Consider the coordinates of a point before and after each move modulo 2.

Define the four corners of the square as A, B, C, and D with coordinates $(0,0)$, $(1,0)$, $(1,1)$, and $(0,1)$ respectively. Exploring with random moves and labelling the points where each peg lands, we may notice that we never need to give a point two different labels. This indicates some sort distinction between the pegs that is invariant and motivates understanding this invariant.

We call a point reachable by a peg starting on point i as a point of type i. Our method of proof will be to show that this definition is well-defined, that is, each point has a unique type. This is equivalent to showing that a peg starting on a point with a given label cannot reach any point that is reachable by a peg starting on a point with a different label. Once this is fact has been established then we see that the last corner of the square cannot be reached as it is of type D yet the initial pegs are of type A, B, and C.

If a peg with position vector $\mathbf{i}$ jumps over a peg with position vector $\mathbf{j}$ then the position vector of the point that the peg lands is $\mathbf{j}+(\mathbf{j}-\mathbf{i})$. By playing out a few moves and labelling the points with a type according to the rule defined previously, we hypothesize that the labels follow a pattern that repeats every two rows and every two columns. This motivates us to consider the coordinates modulo 2 as showing that two numbers are different modulo 2 is sufficient to show the two numbers are different. We apply this to our new position vector in (15).

$$\begin{aligned} & \mathbf{j}+(\mathbf{j}-\mathbf{i}) && \mod 2 \\ \equiv\ & 2\mathbf{j}-\mathbf{i} && \mod 2 \\ \equiv\ & \mathbf{i} && \mod 2 \end{aligned} \tag{15}$$

This shows that the position of a point of type i modulo 2 is preserved. All of the initial pegs had different positions modulo 2, proving that a peg of type i can never land on a point of type $j \neq i$. By our previous comments.

(c) For the starting configuration in figure 20a, characterise (with proof) exactly which lattice points can be reached.

Solution:

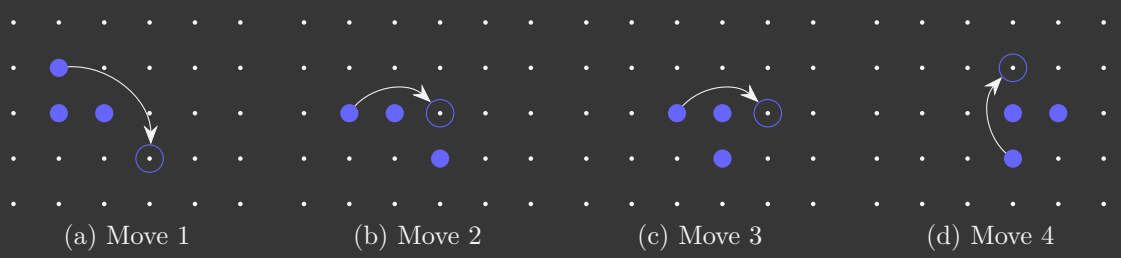
In the solution presented to part b we hypothesized that the pattern of how points repeated every two units in each direction, but never proved this fact. We showed that the pattern could not repeat in any direction by any number not divisible by 2. The pattern does in fact repeat every two units. First we simplify the problem. Suppose that there exists a sequence of moves that results in the original pattern of three pegs being moved to the right by two units. The we make the following observations.

1. There exists a mirror of symmetry that cuts diagonally through the original peg positions, therefore if the three pegs can be moved to the right by two units then they can be moved up by the same sequence performed when mirrored.
2. The moves are reversible so if the pegs can move right by two then they can move left by two, and likewise for moving up and down by two.
3. All 2×2 squares could therefore be reached by the original three pegs.

All that is left to do is come up with such a sequence that moves the initial configuration to the right by two units. In fact, by the symmetries in our argument, finding a sequence that moves the initial configuration in any cardinal direction by two units would be sufficient. This can be done in four moves as follows where we assume the pegs are at (0, 0), (0, 1), and (1, 0), and is presented in figure 21.

1. Move the peg at (0, 1) over the peg at (1, 0) to (2, -1).
2. Move the peg at (0, 0) over the peg at (1, 0) to (2, 0).

3. Move the peg at (1, 0) over the peg at (2, 0) to (3, 0).
4. Move the peg at (2, -1) over the peg at (2, 0) to (2, 1).



(a) Move 1 (b) Move 2 (c) Move 3 (d) Move 4

Figure 21: Moving the initial configuration to the right by two units.

(d) Consider the higher-dimensional analagues of part c. On an n-dimensional infinite lattice there are pegs at the origin and one unit along each of the coordinate axes. Explicitly we have $v_0 = (0, \ldots, 0)$ and $v_k = (0, \ldots, 0, 1, 0, \ldots, 0)$ where the 1 occurs in the kth position for $k = 1, \ldots, n$. Using your answers from parts b and c or otherwise, characterise (with proof) exactly which lattice points can be reached for this starting configuration.

Hints:

1. This is very similar to the earlier parts. This question is only testing whether you are capable of abstracting.
2. Can you move one point at a time?
3. Consider the origin, a point in the cardinal direction you want to move, and another point. This defines a plane and the problem is exactly the same as the two-dimensional variant, expect the first two points need to end up in the same position they started in after any sequence of moves.

Solution:

The proof in part b is independent of dimension[7] so again we see that the coordinates of our points modulo 2 is invariant. This has characterised some points as definitely unreachable, but we still need to prove that the remaining points are reachable. The proof from part c can be modified to work for this higher dimensional configuration.

Pick a direction to move along, let us say along the kth axis, we show that all the other points can be moved by two units in this direction. We look at each v_i where $i \neq k$ and show how to move that point by two units. Looking at the plane passing through the origin, v_k, and v_i, we see we are in exactly the same situation as part c and therefore the same sequence of moves can be applied.

The peg now at $3v_k$ can jump over the peg now at $2v_k$ to land at v_k where it started. Similarly the peg at v_2 can jump over the peg at v_k to land back at the origin. This situation is now identical to the starting configuration, apart from the peg that started at v_i has been moved to $v_i + 2v_k$. We can repeat this for all $i \neq k$, and if the last two "undoing moves" are not done then the origin and v_k are also moved two units right, meaning the whole configuration has been moved. This procedure is actually more complicated than it needs to be, and we will see an even simpler method of moving points in part e.

[7] It even works in infinitely many dimensions.

(e) Consider a three-dimensional infinite lattice with pegs at $(1,0,0)$, $(0,1,0)$, and $(0,0,1)$. How does the characterisation of reachable lattice points differ from part d? Hence conjecture what the characterisation might be for any starting configuration of pegs on an n-dimensional infinite lattice. Can you prove this?

Solution:

We cannot do the same proof as in part d because we no longer have a peg at the origin. The three pegs define the plane and all moves keep the pegs within that plane. This is because the vector that each peg moves by is necessarily contained within the plane by definition of a move. If we restrict the lattice to only the points contained within the same plane as the pegs then we see they form a two-dimensional lattice. The same result as in part c gives the characterisation in this case, but restricted to this new lattice.

Looking towards generalising this to any configuration of points, we extend this observation to get a restriction on what points are reachable. Reachable points must lie within the space that the vectors that connect each pair of points can cover when starting from one of the pegs. More formally, if V is the set of differences between the position vectors of the pegs and v is the position vector of an arbitrary peg, then the set of reachable points must be a subset of $R = \{v + \sum_i^N \alpha_i v_i | \alpha_i \in \mathbb{Z}, v_i \in V, n \in \mathbb{N}\}$[8].

We also know that the coordinates must preserve the parity of the original pegs. Based on the intuition we have built up so far, we conjecture that the points in R that have the same parity as one of the original pegs are the reachable points. To prove this we can emulate the proof from part d. It is sufficient to show that for any initial peg positions v_i and v_j, any other peg can be moved by $2(v_i - v_j)$ while leaving the other pegs unchanged. The factor of two is needed to preserve parity.

We prove this for the simpler case where all pegs have different parity. Without loss of generality, we consider three pegs P_1, P_2, and P_3 with position vectors v_1, v_2, and v_3, and we attempt to move v_3 by $2(v_2 - v_1)$ with the pegs at v_1 and v_2 returning to their original positions. All our moves here are well-defined as a peg cannot land on another peg due to having different parities. If P_3 jumps over P_1 then over P_2 then it initially moves to $v_1 + (v_1 - v_3) = 2v_1 - v_3$, then to $v_2 + (v_2 - (2v_1 - v_3)) = v_3 + 2(v_2 - v_1)$.

The general case where pegs can have the same parity is more complicated, but we can apply a similar idea to the simple case. If we cannot jump peg P_3 over P_1 we can move P_1 or P_2 and try again, let us say we jump P_1 over P_2 and relabel it peg P_1'. We can apply the same method as before but with pegs P_2 and P_1' instead of pegs P_1 and P_2. The vector from P_1 to P_2 is the same as the vector from P_2 to P_1' so P_3 will be moved by the same amount. If this does not work then we can keep jumping pegs P_1 and P_2 over each other until they are far away.

We still need to prove that this process works and is always possible. There are potential problems.

1. Is this process of flipping P_1 and P_2 guaranteed to work after finitely many time steps? That is, could we get stuck in a loop where no matter how many times we flip P_1 and P_2 over each other, there is always some other peg that gets in the way.
2. Can we always flip P_1 and P_2 over each other?

If we have finitely many pegs, let us say n, then the answer to the first statement is yes. After each flip of P_1 and P_2, the result of flipping P_3 over each of P_1 and P_2 must be different. This means that after $2n$ total flips, alternating between P_1 and P_2, we must

have reached some state where all moves involving P_3 were well-defined by the pigeonhole principle. P_1 and P_2 can then retrace their steps and return to their original positions.

For P_1 not to be able to flip over P_2 there must exist some peg, P_4 that is an equal distance away from P_2 as P_1 but in the opposite direction. If this is the case then we can use P_2 and P_4 instead of P_1 and P_2 as they necessarily have the same vector between them.

The situation with infinitely many pegs or pegs on an infinite dimensional lattice is more complicated due to potential issues about convergence and is therefore not covered here. We note that having infintely many pegs does not necessarily make the situation more complicated. Consider a one-dimensional lattice with pegs at even-numbered positions and one peg at an odd-numbered position - here it is clear that all points are reachable.

[8] All finite linear combinations of $v_i \in V$ is referred to as the span of V and is a vector space. $R = v + \operatorname{span} U$ is known as an affine space.